

Integrals involving Fourier basis functions

The below theorem is used everywhere.

Theorem 0.1 (Fourier Integral):

$$\int_0^{2\pi} e^{imx} dx = \begin{cases} 2\pi(m=0) \\ 0 & (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \quad (1)$$

Proof:

$$\int_0^{2\pi} e^{imx} dx = \frac{e^{im \cdot 2\pi} - e^{im \cdot 0}}{im} \quad (2.1)$$

$$= \begin{cases} 2\pi(m=0) \\ 0 & (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \quad (2.2)$$

□

- Integral equations coming from common partial differential equations in $\mathbb{S}^1$ ($\mathbb{R}^2$) often involve singular integrals, derivative of integrals.
- To compute these integrals, exact values of integrals of Fourier basis functions are needed.
- To deal with the singularity while maintaining 2π -periodicity, we need a function which only has singularity at $0(2\pi)$ and is 2π -periodic.
 - $\csc^n t, \cot^n t, n \in \mathbb{N}$: singularities at π as well
 - $\csc^n \frac{t}{2}, n \notin 2\mathbb{N}$: not 2π -periodic
 - $\cot^n \frac{t}{2}, n \in \mathbb{N}$: ok
 - $\csc^{2n} \frac{t}{2}, n \in 2\mathbb{N}$: ok

Definition 0.1 (Cauchy Principle value on a circle): $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall f \in C^{0,\alpha}(0, 2\pi)$

$$\oint_0^{2\pi} f(x) \cot\left(\frac{x-t}{2}\right) dx := \lim_{\varepsilon \rightarrow 0} \left(\int_{\varepsilon}^{2\pi-\varepsilon} f(x+t) \cot\left(\frac{x}{2}\right) dx \right) \quad (3)$$

Definition 0.2 (Hadamard Finite-Part on a circle): $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall n \in \mathbb{N} \setminus \{1\}. \forall f \in C^{n-1,\alpha}(0, 2\pi).$

$$\oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx := \lim_{\varepsilon \rightarrow 0} F_0(\varepsilon) \quad (4)$$

where

$$\forall \varepsilon \in (0, \pi). \int_{\varepsilon}^{2\pi-\varepsilon} f(x+t) \cot^n\left(\frac{x}{2}\right) dx = F_0(\varepsilon) + F_1(\varepsilon) \quad (5.1)$$

$$\lim_{\varepsilon \rightarrow 0} F_0(\varepsilon) < \infty \quad (5.2)$$

$$\lim_{\varepsilon \rightarrow 0} \log\left(4 \sin^2\left(\frac{\varepsilon}{2}\right)\right) \frac{1}{F_1(\varepsilon)} < \infty \quad (5.3)$$

Theorem 0.2 (Hadamard Finite-Part on a circle): $\forall n \in \mathbb{N}. \forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi).$ Let

$$(T_n f)(t) := \oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx \quad (6)$$

then

$$(T_{n+1}f)(t) := -\frac{2}{n} \frac{d}{dt}(T_n f)(t) - (T_{n-1}f)(t) \quad (7)$$

where

$$(T_1 f)(t) := \oint_0^{2\pi} f(x) \cot\left(\frac{x-t}{2}\right) dx \quad (8.1)$$

$$(T_0 f)(t) := \int_0^{2\pi} f(x) dx \quad (8.2)$$

Theorem 0.3: $\forall n \in \mathbb{N}. \forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi)$.

$$\frac{d}{dt} \oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx = \oint_0^{2\pi} f'(x) \cot^n\left(\frac{x-t}{2}\right) dx \quad (9)$$

Proof:

$$\frac{d}{dt} \oint_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) dx = \frac{d}{dt} \oint_0^{2\pi} f(x+t) \cot^n\left(\frac{x}{2}\right) dx \quad (10.1)$$

$$= \frac{d}{dt} \lim_{\varepsilon \rightarrow 0} F(t, \varepsilon) \quad (10.2)$$

$$\stackrel{?}{=} \lim_{\varepsilon \rightarrow 0} \frac{d}{dt} F(t, \varepsilon) \quad (10.3)$$

$$= \oint_0^{2\pi} f'(x) \cot^n\left(\frac{x-t}{2}\right) dx \quad (10.4)$$

□

Theorem 0.4:

$$\oint_0^{2\pi} f'(t)g(t) dt \stackrel{?}{=} - \oint_0^{2\pi} f(t)g'(t) dt \quad (11)$$

Theorem 0.5: Let

$$I_{m,n} = \oint_0^{2\pi} e^{imt} \cot^n\left(\frac{t}{2}\right) dt \quad (12)$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}. \forall n \in \mathbb{N} \setminus \{1\}. I_{m,n} = \frac{2im}{n-1} I_{m,n-1} - I_{m,n-2} \quad (13)$$

with initial values:

$$I_{m,0} = 0, \quad I_{m,1} = 2\pi i \operatorname{sgn}(m) \quad (m \neq 0) \quad (14.1)$$

$$I_{0,0} = 2\pi, \quad I_{0,1} = 0 \quad (14.2)$$

Proof:

- $I_{m,0}, m \in \mathbb{Z}$: Theorem 0.1.
- $I_{0,1} = 0$: Due to the asymmetry of integrand $\cot \frac{t}{2}$.
- $I_{m,1}, m \in \mathbb{Z}$: Follows [1, Lemma 8.23.].
 - $\cot \frac{t}{2} = \frac{\cos \frac{t}{2}}{\sin \frac{t}{2}} = i \frac{e^{\frac{it}{2}} + e^{-\frac{it}{2}}}{e^{\frac{it}{2}} - e^{-\frac{it}{2}}} = i \frac{e^{it} + 1}{e^{it} - 1}$

- $\therefore \forall m \in \mathbb{N}. (e^{imt} - 1) \cot\left(\frac{t}{2}\right) = (e^{it} - 1) \left(\sum_{j=0}^{m-1} e^{ijt} \right) \cot\left(\frac{t}{2}\right) = i(e^{it} + 1) \sum_{j=0}^{m-1} e^{ijt}.$
- $\therefore \forall m \in \mathbb{N}. I_{m,1} - I_{0,1} = 2\pi i$ Integrating both hands and Theorem 0.1
- $\therefore \forall m \in \mathbb{N}. I_{m,1} = 2\pi i$
- $\forall m \in \mathbb{N}, I_{-m,1} = \overline{I_{m,1}} = -2\pi i$
- Recurrence relation: Due to Theorem 0.2 Theorem 0.3 and $(e^{imt})' = ime^{imt}.$

□

Theorem 0.6: Let

$$J_{m,n} = \oint_0^{2\pi} e^{imt} \log\left(4 \sin^2\left(\frac{t}{2}\right)\right) \cot^n\left(\frac{t}{2}\right) dt \quad (15)$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}. \forall n \in \mathbb{N} \setminus \{1\}. J_{m,n} = \frac{2im}{n-1} J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1} I_{m,n} \quad (16)$$

with initial values:

$$J_{m,0} = -\frac{2\pi}{|m|}, \quad J_{m,1} = 2\pi i \operatorname{sgn}(m) \left(2H_{|m|} - \frac{1}{|m|}\right), \quad H_m := \sum_{k=1}^m \frac{1}{k} \quad (m \neq 0) \quad (17.1)$$

$$J_{0,0} = 0, \quad J_{0,1} = 0 \quad (17.2)$$

Proof:

- $J_{m,0}, m \in \mathbb{Z}$: Follows [1, Lemma 8.23.].
- $J_{0,1}$: Due to the asymmetry of integrand $\log(4 \sin^2(\frac{t}{2})) \cot(\frac{t}{2})$.
- $J_{m,1}, m \in \mathbb{Z} \setminus \{0\}$:
 - $L(t) := \log(4 \sin^2(\frac{t}{2})) = -\sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{ikt}}{|k|}$
 - $\forall m \in \mathbb{N}. J_{m,1} = \oint_0^{2\pi} e^{imt} L(t) L'(t) dt = -\frac{im}{2} \int_0^{2\pi} e^{imt} L^2(t) dt =$
 $-\frac{im}{2} \int_0^{2\pi} e^{imt} \left(\sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{ikt}}{|k|} \right) \left(\sum_{l \in \mathbb{Z} \setminus \{0\}} \frac{e^{ilt}}{|l|} \right) dt = (2\pi) \sum_{k,l \in \mathbb{Z} \setminus \{0\}, k+l+m=0} \frac{1}{|k||l|} =$
 $(2\pi) \sum_{k \in \mathbb{Z} \setminus \{0\}, l \in \mathbb{Z}, k+l+m=0} \frac{1}{|k||l|} = (2\pi) \sum_{k \in \mathbb{Z} \setminus \{0, -m\}} \frac{1}{|k||m+k|}$
 - $\forall m \in \mathbb{N}. \sum_{k \in \mathbb{Z} \setminus \{0, -m\}} \frac{1}{|k||m+k|} = \sum_{k=1}^{\infty} \frac{1}{k(m+k)} + \sum_{k=-m+1}^{-\infty} \frac{1}{|k||m+k|} +$
 $\sum_{k=-\infty}^{-m+1} \frac{1}{|k||m+k|} = \sum_{k=1}^{\infty} \frac{1}{k(m+k)} + \sum_{k=1}^{m-1} \frac{1}{k(m-k)} + \sum_{k=1}^{\infty} \frac{1}{k(m+k)} = \frac{2}{m} \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{m+k} \right) +$
 $\frac{1}{m} \sum_{k=1}^{m-1} \left(\frac{1}{k} + \frac{1}{m-k} \right) = \frac{2}{m} \left(\sum_{k=1}^m \frac{1}{k} + \sum_{k=1}^{m-1} \frac{1}{k} \right) = \frac{2}{m} (2H_m - \frac{1}{m})$

□

Theorem 0.7 (Estimates for $I_{m,n}, J_{m,n}$):

1. $\forall m \in \mathbb{Z} \setminus \{0\}. \forall n \in \mathbb{N}_0. |I_{m,n}| \leq 2\pi n |m|^{n-1}$
2. $\forall m \in \mathbb{Z} \setminus \{0\}. \forall n \in \mathbb{N}_0. |J_{m,n}| \leq 2\pi(n+1)^2 |m|^n$

Proof: Proved by induction for n

1. $I_{m,n}$
 - $n = 0$: $|I_{m,0}| = 0$
 - $n = 1$: $|I_{m,1}| = 2\pi$
 - $n > 1$: $|I_{m,n}| = \left| \frac{2im}{n-1} I_{m,n-1} - I_{m,n-2} \right| \leq \frac{2|m|}{n-1} |I_{m,n-1}| + |I_{m,n-2}| \leq \frac{2|m|}{n-1} \cdot 2\pi(n-1) |m|^{n-2} + 2\pi(n-2) |m|^{n-3} \leq 4\pi |m|^{n-1} + 2\pi(n-2) |m|^{n-3} \leq 2\pi n |m|^{n-1}$
2. $J_{m,n}$
 - $n = 0$: $|J_{m,0}| = \frac{2\pi}{|m|} \leq 2\pi$
 - $n = 1$: $|J_{m,1}| = 2\pi \left| 2H_{|m|} - \frac{1}{|m|} \right| \leq 8\pi |m|$

$$\begin{aligned} \bullet \quad n > 1: |J_{m,n}| &= \left| \frac{2im}{n-1} J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1} I_{m,n} \right| \leq \frac{2|m|}{n-1} |J_{m,n-1}| + |J_{m,n-2}| + \\ &\quad \frac{2}{n-1} |I_{m,n}| \leq \frac{2|m|}{n-1} \cdot 4\pi n^2 |m|^{n-1} + 4\pi(n-1)^2 |m|^{n-2} + 4\pi \frac{n}{n-1} |m|^{n-1} \leq 4\pi(4n + \\ &\quad (n-1)^2 + 2) |m|^n = 4\pi(n^2 + 2n + 1) |m|^n = 4\pi(n+1)^2 |m|^n \end{aligned}$$

□

Subspace U_N

Lemma 0.1: $\forall N' \in \mathbb{N}. t_j := \frac{2\pi j}{N'}. \forall m \in \mathbb{Z}. \forall t_{\text{start}} \in [0, 2\pi).$

$$\frac{2\pi}{N'} \sum_{j=0}^{N'-1} e^{im(t_j - t_{\text{start}})} = \begin{cases} 2\pi(m \equiv 0 \pmod{N'}) \\ 0 \quad (\text{otherwise}) \end{cases} \quad (18)$$

Definition 0.3: $U_N := \text{span}(\{e^{imx} \mid m \in \mathbb{Z}, |m| < N\})$

Theorem 0.8 (Trapezoidal Rule for U_N): $\forall N \in \mathbb{N}. N$ -point trapezoidal rule is exact for U_N .

Proof: Lemma 0.1

□

Lemma 0.2 (Trapezoidal Rule for inner product for U_N): Let $N' := \dim U_N = 2N - 1$. Let $t_j := \frac{2\pi j}{N'}$ for $j = 0, \dots, N' - 1$. $\forall f, g \in U_N$.

$$\int_0^{2\pi} f(t)g(t) dt = \frac{2\pi}{N'} \sum_{j=0}^{N'-1} f(t_j)g(t_j) \quad (19)$$

Proof: Since, $f(t)g(t) \in U_{2N-1}$ Theorem 0.8 can be applied with $N' = 2N - 1$ points.

□

Quadratures

Now we consider the way to express discrete Fourier expansion of a function in U_N using the values of the function at $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \dots$

Lemma 0.3: $\forall N \in \mathbb{N}. \forall f \in U_N. N := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$f(x) = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left(\frac{1}{N'} \sum_{|m| < N} e^{-im(t_j + t_{\text{start}})} e^{imx} \right) \quad (20)$$

Proof:

$$f(x) = \sum_{|m| < N} \int_0^{2\pi} f(t) \frac{e^{-imt}}{\sqrt{2\pi}} dt \cdot \frac{e^{imx}}{\sqrt{2\pi}} \quad (21.1)$$

$$\stackrel{\text{Lemma 0.2}}{=} \frac{1}{2\pi} \sum_{|m| < N} \frac{2\pi}{N'} \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) e^{-im(t_j + t_{\text{start}})} \cdot e^{imx} \quad (21.2)$$

$$= \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left(\frac{1}{N'} \sum_{|m| < N} e^{-im(t_j + t_{\text{start}})} e^{imx} \right) \quad (21.3)$$

□

Definition 0.4 (Lagrange basis): $\forall N \in \mathbb{N}. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$L_j^{t_{\text{start}}}(x) := \frac{1}{N'} \sum_{|m| < N} e^{-im(t_j + t_{\text{start}})} e^{imx} \quad (22)$$

Theorem 0.9: $\forall N \in \mathbb{N}. N' := 2N - 1. \forall t_{\text{start}}, t'_{\text{start}} \in [0, 2\pi). \forall i, j \in \{0, \dots, N' - 1\}. t_i := \frac{2\pi i}{N'}. t_j := \frac{2\pi j}{N'}.$

$$\left\langle L_i^{t_{\text{start}}}, L_j^{t'_{\text{start}}} \right\rangle = \frac{2\pi}{N'^2} \sum_{|m| < N} e^{im(t_j+t'_{\text{start}}-t_i-t_{\text{start}})} \quad (23)$$

Proof:

$$\left\langle L_i^{t_{\text{start}}}, L_j^{t'_{\text{start}}} \right\rangle = \int_0^{2\pi} L_i^{t_{\text{start}}}(t) \overline{L_j^{t'_{\text{start}}}(t)} dt \quad (24.1)$$

$$= \int_0^{2\pi} \left(\frac{1}{N'} \sum_{|m| < N} e^{-im(t_i+t_{\text{start}})} e^{imt} \right) \left(\frac{1}{N'} \sum_{|n| < N} e^{in(t_j+t'_{\text{start}})} e^{-int} \right) dt \quad (24.2)$$

$$= \frac{1}{N'^2} \sum_{|m| < N} \sum_{|n| < N} e^{-im(t_i+t_{\text{start}})} e^{in(t_j+t'_{\text{start}})} \int_0^{2\pi} e^{i(m-n)t} dt \quad (24.3)$$

$$= \frac{1}{N'^2} \sum_{|m| < N} \sum_{|n| < N} e^{-im(t_i+t_{\text{start}})} e^{in(t_j+t'_{\text{start}})} \delta_{m,n} 2\pi \quad (24.4)$$

$$= \frac{2\pi}{N'^2} \sum_{|m| < N} e^{im(t_j+t'_{\text{start}}-t_i-t_{\text{start}})} \quad (24.5)$$

□

Combining the above lemma with the exact integral values of Fourier basis functions, we can exactly compute the integrals of functions in U_N multiplied by $\cot^n(\frac{t}{2})$ or $\log(4 \sin^2(\frac{t}{2})) \cot^n(\frac{t}{2})$, by using only the values of the function at $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \dots$

Theorem 0.10 (Generalized Garrick–Wittich quadrature for U_N): $\forall n \in \mathbb{N}_0. \forall N \in \mathbb{N}. \forall f \in U_N. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$\oint_0^{2\pi} f(t) \cot^n\left(\frac{t}{2}\right) dt = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) P_j^{N', n} \quad (25)$$

$$P_j^{N', n, t_{\text{start}}} := \frac{1}{N'} \sum_{|m| < N} I_{m,n} e^{-im(t_j+t_{\text{start}})} \quad (26)$$

Theorem 0.11 (Generalized Kussmaul–Martensen (Kress) quadrature for U_N): $\forall n \in \mathbb{N}_0. \forall N \in \mathbb{N}. \forall f \in U_N. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi).$

$$\oint_0^{2\pi} f(t) \log\left(4 \sin^2\left(\frac{t}{2}\right)\right) \cot^n\left(\frac{t}{2}\right) dt = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) Q_j^{N', n} \quad (27)$$

$$Q_j^{N', n, t_{\text{start}}} := \frac{1}{N'} \sum_{|m| < N} J_{m,n} e^{-im(t_j+t_{\text{start}})} \quad (28)$$

Error estimates

Integral Equations

Theorem 0.12: X : Banach space, $\forall A \in B(X). \forall T \in B^{\text{inv}}(X). \|T^{-1}(T - A)\| < 1 \implies A \in B^{\text{inv}}(X)$

Proof: Neumann series implies $(I - (T^{-1}(A - T))) \in B^{\text{inv}}(X)$ and

$$\|A^{-1}\| = \left\| (I - (T^{-1}(T - A)))^{-1} \right\| \leq \frac{1}{1 - \|T^{-1}(T - A)\|} \quad (29)$$

□

Definition 0.5:

$$I_N : \rightarrow U_N, f \mapsto \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) L_{t_{\text{start}}}(x) \quad (30)$$

Theorem 0.13: Let φ solution of

$$\varphi(s) + \sum_{n=0}^M \int_0^{2\pi} K_n(s, t) \cot^n\left(\frac{s-t}{2}\right) \varphi(t) dt \quad (31.1)$$

$$+ \sum_{n=0}^M \int_0^{2\pi} L_n(s, t) \log\left(4 \sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right) \varphi(t) dt = f(s) \quad (31.2)$$

and $\tilde{\varphi}^{(N)} \in U_N$ solution of

$$\tilde{\varphi}^{(N)}(s) + I_N \sum_{n=0}^M \int_0^{2\pi} K_n(s, t) \cot^n\left(\frac{s-t}{2}\right) \tilde{\varphi}^{(N)}(t) dt \quad (32.1)$$

$$+ I_N \sum_{n=0}^M \int_0^{2\pi} L_n(s, t) \log\left(4 \sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right) \tilde{\varphi}^{(N)}(t) dt = I_N f(s) \quad (32.2)$$

assume that Equation 31 is solvable. Then

1. $\exists N_s \in \mathbb{N}. \|T - A\| < \|T^{-1}\|^{-1}$
2. $\forall N_s$ [satisfies above condition]. $\forall N \geq N_s$. Equation 32 has a unique solution

Theorem 0.14: Let $\{\tilde{\varphi}_j^{(N)}\}_{j=0}^{N'-1}$ solution of

$$\sum_{j=0}^{N'-1} L_j^{t'_{\text{start}}}(t_i + t_{\text{start}}) \tilde{\varphi}_j^{(N)} + \sum_{n=0}^M \sum_{j=0}^{N'-1} K_n(t_i + t_{\text{start}}, t_j + t'_{\text{start}}) P_j^{N', n, t'_{\text{start}}} \tilde{\varphi}_j^{(N)} \quad (33.1)$$

$$+ \sum_{n=0}^M \sum_{j=0}^{N'-1} L_n(t_i + t_{\text{start}}, t_j + t'_{\text{start}}) Q_j^{N', n, t'_{\text{start}}} \tilde{\varphi}_j^{(N)} = f(t_i + t_{\text{start}}) \quad (33.2)$$

Then

$$\tilde{\varphi}^{(N)} = \sum_{j=0}^{N'-1} \tilde{\varphi}_j^{(N)} L_j^{t'_{\text{start}}} \quad (34)$$

Proof: Evaluate Equation 32 at $s = t_i + t_{\text{start}}$.

$$\sum_{j=0}^{N'-1} L_j^{t'_{\text{start}}}(t_i + t_{\text{start}}) \tilde{\varphi}_j^{(N)} + \sum_{n=0}^M \sum_{j=0}^{N'-1} K_n(t_i + t_{\text{start}}, t_j + t'_{\text{start}}) P_j^{N', n, t'_{\text{start}}} \tilde{\varphi}_j^{(N)} \quad (35.1)$$

$$+ \sum_{n=0}^M \sum_{j=0}^{N'-1} L_n(t_i + t_{\text{start}}, t_j + t'_{\text{start}}) Q_j^{N', n, t'_{\text{start}}} \tilde{\varphi}_j^{(N)} = f(t_i + t_{\text{start}}) \quad (35.2)$$

□

Bibliography

- [1] R. Kress, *Linear Integral Equations*, vol. 82. in Applied Mathematical Sciences, vol. 82. New York, NY: Springer New York, 2014. doi: 10.1007/978-1-4614-9593-2.